Michael Gruenstaeudl (Grünstäudl), PhD

Fort Hays State University, 600 Park Street, Hays, KS 67601, USA

m_gruenstaeudl@fhsu.edu | gruenstaeudl-lab.com | 0000-0002-1666-1773

Education and Professional Positions

Education

Habilitation in Bioinformatics and Botany — Freie Universität Berlin, Germany — 2023

Habilitation thesis: "Development and application of bioinformatic tools toward process automation in plant phylogenetics"

Ph.D. in Plant Biology — University of Texas at Austin, USA — 2013

M.Sc. in Plant Biology — University of Vienna, Austria — 2007

Professional Positions

Assistant Professor (Tenure-Track) — Fort Hays State University, USA — 2023–ongoing
Dept. Biological Sciences

Postdoctoral Researcher — Freie Universität Berlin, Germany — 2015–2022
Dept. Biology, Chemistry, Pharmacy

Postdoctoral Researcher — Ohio State University, USA — 2014–2015
Dept. Evolution, Ecology & Organismal Biology

Research

Research Interests

Phylogenomics and bioinformatics of evolutionary genomic data • Computational analysis of large-scale sequence datasets • Systematics and evolutionary biology of plants, algae, and cyanobacteria

Research Funding / Grants

[8] M Gruenstaeudl. *Genome sequencing of bloom-forming cyanobacteria in western Kansas*. Data science core / PUI grant. Single PI; Grant no. P20GM103418/GR00848; Duration: 2025–ongoing; Amount: **$ 33,210**. Funder: Kansas INBRE, National Institutes of Health, USA, 2025.

[7] M Gruenstaeudl. *Implementing a course-based undergraduate research experience to train students in identifying bloom-forming cyanobacteria using state-of-the-art genetic techniques*. Research grant. Single PI; Grant no. GR00807; Duration: 2025–ongoing; Amount: **$ 24,059**. Funder: Kansas Water Office, Fort Hays State University, USA, 2025.

[6] M Gruenstaeudl. *Enabling data quality assessment of organelle genomes archived on GenBank through novel open-source software tools*. R01 research grant. Single PI; Grant no. R01LM014506; Duration: 2024–ongoing; Amount: **$ 239,206**. Funder: NLM, National Institutes of Health, USA, 2024. url: https://reporter.nih.gov/search/SLep2Yj6r0WWAVF8ECi_HA/project-details/10858197.

[5] TL Phelps-Durr, M Gruenstaeudl, LE Patrick, SD Elzay, and NK Lazerus. *Uncovering change catalysts: Examining Community of Practice members as they leverage individual course improvements to achieve systemic departmental change*. IUSE: EDU research grant. Co-PI; Grant no. 2417083; DURATION: 2024–ongoing; AMOUNT: **$ 385,971**. FUNDER: National Science Foundation, USA, 2024. URL: https://www.nsf.gov/awardsearch/show-award/?AWD_ID=2417083.

[4] M Gruenstaeudl. *Data mining of organellar genomes using bioinformatics–A powerful method to understand the quality of publicly accessible organellar sequence records*. Start-up grant. Single PI; Grant no. P20GM103418/GR509061; DURATION: 2023–2024; AMOUNT: **$ 24,765**. FUNDER: Kansas INBRE, National Institutes of Health, USA, 2023.

[3] M Gruenstaeudl. *Plastid phylogenomics of water lilies, with a focus on partitioning strategies and the development of bioinformatic tools*. Research grant (Sachbeihilfe). Single PI; Grant no. 418670221; DURATION: 2018–2022; AMOUNT: **€ 69,360**. FUNDER: Deutsche Forschungsgemeinschaft, Germany, 2018. URL: https://gepris.dfg.de/gepris/projekt/418670221?language=en.

[2] M Gruenstaeudl. *Plastid phylogenomics of water lilies (Nymphaeaceae) and early-diverging sunflowers (Barnadesioideae)*. Seed grant. Single PI; Grant no. 21224600; DURATION: 2016–2018; AMOUNT: **€ 11,470**. FUNDER: Forschungskommission der Freien Universität Berlin, Germany, 2016.

[1] M Gruenstaeudl. *Graduate Research Fellowship*. Research fellowship. Single PI; Grant no. F816842; DURATION: 2011–2012; AMOUNT: **$ 26,772**. FUNDER: University of Texas at Austin, USA, 2011.

Publications

Graduate and undergraduate student mentees are <u>underlined</u>

[27] <u>N Jenke</u>, <u>Smith, GM</u>, <u>Magar Thapa, B</u>, and **Gruenstaeudl, M**. “Variation of and associations with the depth and evenness of sequencing coverage in archived plastid genomes”. In: *Scientific Reports* 15 (2025). https://doi.org/10.1038/s41598-025-11568-9, p. 26294.

[26] <u>JA Roestel</u>, JH Wiersema, RK Jansen, T Borsch, and **Gruenstaeudl, M**. “On the importance of sequence alignment inspections in plastid phylogenomics – an example from revisiting the relationships of the water-lilies”. In: *Cladistics* 40 (2024). https://doi.org/10.1111/cla.12584, pp. 469–495.

[25] <u>E Giorgashvili</u>, K Reichel, <u>C Caswara</u>, V Kerimov, T Borsch, and **Gruenstaeudl, M**. “Software choice and sequencing coverage can impact plastid genome assembly – A case study in the narrow endemic *Calligonum bakuense*”. In: *Frontiers in Plant Science* 13 (2022). https://doi.org/10.3389/fpls.2022.779830, p. 779830.

[24] <u>B Escobari</u>, T Borsch, TS Quedensley, and **Gruenstaeudl, M**. “Plastid phylogenomics of the Gynoxoid group (Senecioneae, Asteraceae) highlights the importance of motif-based sequence alignment amid low genetic distances”. In: *American Journal of Botany* 108 (2021). https://doi.org/10.1002/ajb2.1775, pp. 2235–2256.

[23] <u>Mehl, T</u> and **Gruenstaeudl, M**. “airpg: Automatically accessing the inverted repeats of archived plastid genomes”. In: *BMC Bioinformatics* 22 (2021). https://doi.org/10.1186/s12859-021-04309-y, p. 413.

[22] I Duran, A Marrero, F Msanda, C Harrouni, **Gruenstaeudl, M**, J Patino, J Caujape-Castells, and C Garcia-Verdugo. “Iconic, threatened, but largely unknown: Biogeography of the Macaronesian dragon trees (*Dracaena* spp.) as inferred from plastid DNA markers”. In: *Taxon* 69 (2020). doi: https://doi.org/10.1002/tax.12215, pp. 217–233.

[21] **Gruenstaeudl, M**. “annonex2embl: automatic preparation of annotated DNA sequences for bulk submissions to ENA”. In: *Bioinformatics* 21 (2020). doi: https://doi.org/10.1093/bioinformatics/btaa209, p. 207.

[20] **Gruenstaeudl, M** and Jenke, N. "PACVr: Plastome Assembly Coverage Visualization in R". In: *BMC Bioinformatics* 36 (2020). doi: https://doi.org/10.1186/s12859-020-3475-0, pp. 3841–3848.

[19] **Gruenstaeudl, M**. "Why the monophyly of Nymphaeaceae currently remains indeterminate: An assessment based on gene-wise plastid phylogenomics". In: *Plant Systematics and Evolution* 305 (2019). doi: https://doi.org/10.1007/s00606-019-01610-5, pp. 827–836.

[18] **Gruenstaeudl, M** and Hartmaring, Y. "EMBL2checklists: A Python package to facilitate the user-friendly submission of plant and fungal DNA barcoding sequences to ENA". In: *PLoS ONE* 14 (2019). doi: https://doi.org/10.1371/journal.pone.0210347, e0210347.

[17] A Szukala, N Korotkova, **Gruenstaeudl, M**, AN Sennikov, GA Lazkov, SA Litvinskaya, SA Gabrielian, T Borsch, and E von Raab-Straube. "Phylogeny of the Eurasian genus *Jurinea* (Asteraceae: Cardueae): Support for a monophyletic genus concept and a first hypothesis on overall species relationships". In: *Taxon* 68 (2019). doi: https://doi.org/10.1002/tax.12027, pp. 112–131.

[16] V Di Vincenzo, **Gruenstaeudl, M**, L Nauheimer, M Wondafrash, P Kamau, S Demissew, and T Borsch. "Evolutionary diversification of the African achyranthoid clade (Amaranthaceae) in the context of sterile flower evolution and epizoochory". In: *Annals of Botany* 122 (2018). doi: https://doi.org/10.1093/aob/mcy055, pp. 69–85.

[15] **Gruenstaeudl, M**, N Gerschler, and T Borsch. "Bioinformatic workflows for generating complete plastid genome sequences - An example from *Cabomba* (Cabombaceae) in the context of the phylogenomic analysis of the water-lily clade". In: *Life* 8 (2018). doi: https://doi.org/10.3390/life8030025, p. 25.

[14] TS Quedensley, **Gruenstaeudl, M**, and RK Jansen. "Phylogenetic relationships of the Mexican tussilaginoid genera (Asteraceae: Senecioneae)". In: *Journal of the Botanical Research Institute of Texas* 12 (2018), pp. 481–498. ISSN: 1934-5259.

[13] **Gruenstaeudl, M**, BC Carstens, A Santos-Guerra, and RK Jansen. "Statistical hybrid detection and the inference of ancestral distribution areas in *Tolpis* (Asteraceae)". In: *Biological Journal of the Linnean Society* 121 (2017). doi: https://doi.org/10.1093/biolinnean/blw014, pp. 133–149.

[12] **Gruenstaeudl, M**, L Nauheimer, and T Borsch. "Plastid genome structure and phylogenomics of Nymphaeales: Conserved gene order and new insights into relationships". In: *Plant Systematics and Evolution* 303 (2017). doi: https://doi.org/10.1007/s00606-017-1436-5, pp. 1251–1270.

[11] N Korotkova, G Parolly, A Khachatryan, L Ghulikyan, H Sargsyan, J Akopian, T Borsch, and **Gruenstaeudl, M**. "Towards resolving the evolutionary history of Caucasian pears (*Pyrus*, Rosaceae) - Phylogenetic relationships, divergence times and leaf trait evolution". In: *Journal of Systematics and Evolution* 56 (2017). doi: https://doi.org/10.1111/jse.12276, pp. 35–47.

[10] E Maharramova, I Huseynova, S Kolbaia, **Gruenstaeudl, M**, T Borsch, and LAH Muller. "Phylogeography and population genetics of the riparian relict tree *Pterocarya fraxinifolia* (Juglandaceae) in the South Caucasus". In: *Systematics and Biodiversity* 16 (2017). doi: https://doi.org/10.1080/14772000.2017.1333540, pp. 14–27.

[9] BC Carstens, **Gruenstaeudl, M**, and NM Reid. "Community trees: Identifying codiversification in the Paramo dipteran community". In: *Evolution* 70 (2016). doi: https://doi.org/10.1111/evo.12916, pp. 1080–1093.

[8] **Gruenstaeudl, M**. "WARACS: Wrappers to automate the reconstruction of ancestral character states". In: *Applications in Plant Sciences* 4 (2016). doi: https://doi.org/10.3732/apps.1500120, p. 1500120.

[7] **Gruenstaeudl, M**, NM Reid, GL Wheeler, and BC Carstens. "Posterior predictive checks of coalescent models: P2C2M, an R package". In: *Molecular Ecology Resources* 16 (2015). doi: https://doi.org/10.1111/1755-0998.12435, pp. 193–205.

[6] **Gruenstaeudl, M**, CV Hawkes, A Santos-Guerra, and RK Jansen. "Preliminary investigations of correlated diversification between plants and their associated arbuscular mycorrhizal fungi in Macaronesia". In: *Proceedings of the Amurga International Conferences on Island Biodiversity 2011*. Ed. by J Caujape-Castells, G Nieto-Feliner, and JM Fernandez-Palacios. Las Palmas, Spain: Fundacion Canaria Amurga Maspalomas, 2013, pp. 146–153. ISBN: 978-84-616-7394-0.

[5] **Gruenstaeudl, M**, A Santos-Guerra, CV Hawkes, and RK Jansen. "Molecular survey of arbuscular mycorrhizal fungi associated with *Tolpis* on three Canarian islands (Asteraceae)". In: *Vieraea* 41 (2013). doi: http://dx.doi.org/10.31939/vieraea.2013.41.17, pp. 233–252. ISSN: 0210-945X.

[4] **Gruenstaeudl, M**, A Santos-Guerra, and RK Jansen. "Phylogenetic analyses of *Tolpis* Adans. (Asteraceae) reveal patterns of adaptive radiation, multiple colonization and interspecific hybridization". In: *Cladistics* 29 (2013). doi: https://doi.org/10.1111/cla.12005, pp. 416–434.

[3] V Funk, A Anderberg, B Baldwin, R Bayer, J Bonifacino, I Breitwieser, L Brouillet, R Carbajal, R Chan, A Coutinho, D Crawford, J Crisci, M Dillon, S Freire, M Galbany Casals, N Garcia-Jacas, B Gemeinholzer, **Gruenstaeudl, M**, HW Lack, and L Watson. "Compositae metatrees: the next generation". In: *Systematics, Evolution and Biogeography of the Compositae*. Ed. by VA Funk, A Susanna, TF Stuessy, and R Bayer. Vienna, Austria: International Association For Plant Taxonomy (IAPT), 2009, pp. 747–777. ISBN: 978-39-501-7543-1.

[2] **Gruenstaeudl, M**, E Urtubey, RK Jansen, R Samuel, MHJ Barfuss, and TF Stuessy. "Phylogeny of Barnadesioideae (Asteraceae) inferred from DNA sequence data and morphology". In: *Molecular Phylogenetics and Evolution* 51 (2009). doi: https://doi.org/10.1016/j.ympev.2009.01.023, pp. 572–587.

[1] TF Stuessy, E Urtubey, and **Gruenstaeudl, M**. "Barnadesieae (Barnadesioideae)". In: *Systematics, Evolution and Biogeography of the Compositae*. Ed. by V.A. Funk, A. Susanna, T.F. Stuessy, and R. Bayer. Vienna, Austria: IAPT, 2009, pp. 215–228. ISBN: 978-39-501-7543-1.

Scientific/Conference Presentations

[30] M Gruenstaeudl. *A snapshot of the quality and reproducibility of GenBank-archived organellar genomes in 2026*. Conference presentation, NISBRE 2026, Bethesda, Maryland, USA, 2026.

[29] M Gruenstaeudl. *Assembly quality and sequencing coverage variation in organellar and cyanobacterial genomes on GenBank: Implications for phylogenetic research*. Invited seminar, Universität Wien, Vienna, Austria, 2026.

[28] M Gruenstaeudl. *From cyanobacteria to chloroplasts: New ways to evaluate plastid and bacterial genome quality using bioinformatics*. Invited seminar, Pittsburg State University, Kansas, USA, 2026.

[27] M Gruenstaeudl. *Moderator of first trainee presentation section*. Session moderator, K-INBRE Symposium 2026, Overland Park, USA, 2026.

[26] M Gruenstaeudl. *Learning from every angle: 360-degree cameras transform blended genetics laboratory instruction*. Conference presentation, OLC Accelerate 2025, Orlando, Florida, USA, 2025.

[25] M Gruenstaeudl. *Novel bioinformatic methods for assessing organelle and cyanobacterial genome quality*. Invited seminar, Universität Regensburg, Regensburg, Germany, 2025.

[24] M Gruenstaeudl. *Developing bioinformatic tools for the data mining and quality control of archived organellar genomes: An ideal setting for training the next generation of bioinformaticians*. Conference presentation, Austrian Bioinformatics Workshop 2024, Graz, Austria, 2024.

[23] M Gruenstaeudl. *Strategies to improve alignment accuracy in large-scale plant phylogenomic analyses*. Conference presentation, 20th International Botanical Congress, Madrid, Spain, 2024.

[22] M Gruenstaeudl. *Using genomic data mining and bioinformatic software development to assess data quality of organellar genomes on GenBank*. Invited seminar, Wichita State University, Kansas, USA, 2024.

[21] M Gruenstaeudl. *Assessing data quality among archived plastid genomes via novel software tools*. Conference presentation (online), Botanical Society of America Conference, Connecticut, USA, 2021.

[20] M Gruenstaeudl. *Assessing sequence coverage and inverted repeat annotations among plastid genomes*. Workshop organizer (online), Botanical Society of America Conference, Connecticut, USA, 2021.

[19] M Gruenstaeudl. *Exploring data quality among accumulated plastid genomes as a prerequisite for barcoding initiatives using whole plastomes*. Conference presentation (online), 19. Österreichische Botanik-Tagung, Krems, Austria, 2021.

[18] M Gruenstaeudl. *On the need for quality control in the generation and analysis of complete plastid genomes*. Conference presentation (online), Deutsche Botanische Gesellschaft, Oldenburg, Germany, 2021.

[17] M Gruenstaeudl. *The bioinformatic application of complete plastid genomes as plant DNA barcodes*. Conference presentation (online), Barcode of Life Initiative, Vienna, Austria, 2020.

[16] M Gruenstaeudl. *Bioinformatic strategies to submit DNA sequences to public sequence repositories*. Workshop organizer, Gesellschaft für Biologische Systematik, Munich, Germany, 2019.

[15] M Gruenstaeudl. *The monophyly of the water lilies remains uncertain*. Conference presentation, Gesellschaft für Biologische Systematik, Munich, Germany, 2019.

[14] M Gruenstaeudl. *Bioinformatic tools for the submission of DNA sequences to public sequence databases*. Conference presentation, Deutsche Botanische Gesellschaft, Klagenfurt, Austria, 2018.

[13] M Gruenstaeudl. *Computational strategies for the selection of data partitioning schemes in phylogenomics*. Conference presentation, Gesellschaft für Biologische Systematik, Vienna, Austria, 2018.

[12] M Gruenstaeudl. *Phylogenomics done properly: How to partition your data*. Workshop organizer, Gesellschaft für Biologische Systematik, Vienna, Austria, 2018.

[11] M Gruenstaeudl. *Plastid phylogenomics of Nymphaeaceae and development of bioinformatic tools*. Conference presentation, Genomics in Biodiversity Research, Berlin, Germany, 2017.

[10] M Gruenstaeudl. *Statistical hybrid detection and its impact on inferring ancestral distribution areas*. Conference presentation, Dahlem Center of Plant Sciences, Berlin, Germany, 2016.

[9] M Gruenstaeudl. *Model-based inferences in plant systematics: Examples from a diverse toolbox*. Invited seminar, Universität Leipzig, Leipzig, Germany, 2015.

[8] M Gruenstaeudl. *Novel bioinformatic methods in model-based inference of plant phylogeography*. Conference presentation, VISCEA Ecology & Evolution Conference, Vienna, Austria, 2015.

[7] M Gruenstaeudl. *Hybrid taxa can mislead phylogeographic analyses in Tolpis (Asteraceae)*. Conference presentation, Society for the Study of Evolution Conference, USA, 2014.

[6] M Gruenstaeudl. *Analyzing the diversification of plants and AM fungi in the age of next-generation sequencing*. Invited seminar, University of Wageningen, The Netherlands, 2011.

[5] M Gruenstaeudl. *Correlated diversification between arbuscular mycorrhizal fungi and their plant hosts*. Conference presentation, 9th International Mycological Conference, Edinburgh, UK, 2010.

[4] M Gruenstaeudl. *Phylogenetic investigations of correlated diversification between vascular plants and their associated AM fungi*. Conference presentation, Flora Macaronesia International Symposium, Azores, Portugal, 2010.

[3] M Gruenstaeudl. *Time and sequence of dispersal events in the Macaronesian island lineage Tolpis*. Conference presentation, Botanical Society of America Conference, Utah, USA, 2009.

[2] M Gruenstaeudl. *Phylogenetic relationships in Tolpis (Asteraceae) and the utility of ETS sequence variation*. Conference presentation, Botany 2008 Conference, Vancouver, Canada, 2008.

[1] M Gruenstaeudl. *Phylogenetic relationships in Barnadesioideae revisited*. Conference presentation, Botany & Plant Biology 2007 Conference, Chicago, USA, 2007.

Honors & Awards for Research

Outstanding Research and Scholarly Activity Award (university-wide)	Fort Hays State University	2025
Outstanding Scholarship Award (Werth College of Science, Technology, and Mathematics)	Fort Hays State University	2025
Scholarship of excellence	Land Niederösterreich	2012
Graduate student research award	American Society of Plant Taxonomists	2011
Graduate student research award	Mycological Society of America	2010
Graduate student research award	Botanical Society of America	2010

Teaching

List of University Courses Taught

Assistant Professor at Fort Hays State Univ.

[66] **Genetics**. *Lectures*. BIOL325, 3 credit hours. Spring. 2026.

[65] **Genetics**. *Laboratories*. BIOL325, 1 credit hour. Spring. 2026.

[64] **Topics in Biology: Crafting Scientific Presentations**. *Graduate course*. BIOL607/G, 2 credit hours. Spring. 2026.

[63] **Topics in Biology: Molecular Biology**. *Graduate course*. BIOL607/G, 4 credit hours. Spring. 2026.

[62] **Botany**. *Lectures*. BIOL250, 3 credit hours. Spring. 2025.

[61] **Botany**. *Laboratories*. BIOL250L, 1 credit hour. Spring. 2025.

[60] **Genetics**. *Lectures*. BIOL325, 3 credit hours. Fall. 2025.

[59] **Genetics**. *Laboratories*. BIOL325, 1 credit hour. Fall. 2025.

[58] **Plant Anatomy**. *Lectures*. BIOL330, 3 credit hours. Fall. 2025.

[57] **Plant Anatomy**. *Laboratories*. BIOL330L, 1 credit hour. Fall. 2025.

[56] **Principles of Biology**. *Lectures*. BIOL180, 3 credit hours. Spring. 2025.

[55] **Topics in Biology: Bioinformatics**. *Graduate course*. BIOL607/G, 3 credit hours. Fall. 2025.

[54] **Genetics**. *Lectures*. BIOL325, 3 credit hours. Spring. 2024.

[53] **Genetics**. *Lectures*. BIOL325, 3 credit hours. Fall. 2024.

[52] **Genetics**. *Laboratories*. BIOL325, 1 credit hour. Spring. 2024.

[51] **Genetics**. *Laboratories*. BIOL325, 1 credit hour. Fall. 2024.

[50] **Principles of Biology**. *Lectures*. BIOL180, 3 credit hours. Fall. 2024.

[49] **Botany**. *Lectures*. BIOL250, 3 credit hours. Fall. 2023.

[48] **Botany**. *Laboratories*. BIOL250L, 1 credit hour. Fall. 2023.

[47] **Genetics**. *Lectures*. BIOL325, 3 credit hours. Spring. 2023.

[46] **Genetics**. *Laboratories*. BIOL325, 1 credit hour. Spring. 2023.

[45] **Principles of Biology**. *Lectures*. BIOL180, 3 credit hours. Spring. 2023.

[44] **Principles of Biology**. *Lectures*. BIOL180, 3 credit hours. Fall. 2023.

[43] **Readings in Biology: The Impact of AI on Biology and Medicine**. *Graduate course*. BIOL482/BIOL882, 2 credit hours. Fall. 2023.

[42] **Topics in Biology: Bioinformatics**. *Graduate course*. BIOL607/G, 3 credit hours. Fall. 2023.

[41] **Topics in Biology: Molecular Biology**. *Graduate course*. BIOL607/G, 4 credit hours. Fall. 2023.

Lecturer at the Freie Universität Berlin

[40] **Botany & Microbiology for Biochemists**. *Lectures*. Course no. 23700. Fall (online). 2021.

[39] **Botany & Plant Physiology for Veterinary Sciences**. *Lectures*. Course no. 23760b-c. Fall (online). 2021.

[38] **Genetics & Genomics**. *Lectures*. Course no. 23771a. Fall (online). 2021.

[37] **Genetics & Genomics**. *Laboratories*. Course no. 23771b. Fall (online). 2021.

[36] **Introduction to Botany & Biodiversity**. *Lectures*. Course no. 23106. Fall (online). 2021.

[35] **Introduction to Botany & Biodiversity**. *Laboratories*. Course no. 23108a-e. Fall (online). 2021.

[34] **Topics in Biology: Evolution**. *Graduate seminar*. Course no. 23653. Spring (online). 2021.

[33] **Topics in Biology: Evolution**. *Graduate laboratories*. Course no. 23654a-b. Spring (online). 2021.

[32] **Botany & Microbiology for Biochemists**. *Lectures*. Course no. 23700. Fall (online). 2020.

[31] **Botany & Plant Physiology for Veterinary Sciences**. *Lectures*. Course no. 23760b-c. Fall (online). 2020.

[30] **Genetics & Genomics**. *Lectures*. Course no. 23771a. Fall (online). 2020.

[29] **Genetics & Genomics**. *Laboratories*. Course no. 23771b. Fall (online). 2020.

[28] **Introduction to Botany & Biodiversity**. *Lectures*. Course no. 23106. Fall (online). 2020.

[27] **Introduction to Botany & Biodiversity**. *Laboratories*. Course no. 23108a-e. Fall (online). 2020.

[26] **Topics in Biology: Evolution**. *Graduate seminar*. Course no. 23653. Spring (online). 2020.

[25] **Topics in Biology: Evolution**. *Graduate laboratories*. Course no. 23654a-b. Spring (online). 2020.

[24] **Botany & Microbiology for Biochemists**. *Lectures*. Course no. 23700. Fall. 2019.

[23] **Botany & Plant Physiology for Veterinary Sciences**. *Lectures*. Course no. 23760b-c. Fall. 2019.

[22] **Genetics & Genomics**. *Lectures*. Course no. 23771a. Fall. 2019.

[21] **Genetics & Genomics**. *Laboratories*. Course no. 23771b. Fall. 2019.

[20] **Introduction to Botany & Biodiversity**. *Lectures*. Course no. 23106. Fall. 2019.

[19] **Introduction to Botany & Biodiversity**. *Laboratories*. Course no. 23108a-e. Fall. 2019.

[18] **Research in Bioinformatics**. *Laboratories*. Course no. 19400432. Spring. 2019.

[17] **Topics in Biology: Evolution**. *Graduate seminar*. Course no. 23653. Spring. 2019.

[16] **Topics in Biology: Evolution**. *Graduate laboratories*. Course no. 23654a-b. Spring. 2019.

[15] **Botany & Microbiology for Biochemists**. *Lectures*. Course no. 23700. Fall. 2018.

[14] **Botany & Plant Physiology for Veterinary Sciences**. *Lectures*. Course no. 23760b-c. Fall. 2018.

[13] **Current Topics in Plant Systematics & Evolution**. *Graduate seminar*. Course no. 23815. Fall. 2018.

[12] **Genetics & Genomics**. *Lectures*. Course no. 23771a. Fall. 2018.

[11] **Genetics & Genomics**. *Laboratories*. Course no. 23771b. Fall. 2018.

[10] **Introduction to Botany & Biodiversity**. *Lectures*. Course no. 23106. Fall. 2018.

[9] **Introduction to Botany & Biodiversity**. *Laboratories*. Course no. 23108a-e. Fall. 2018.

[8] **Botany & Microbiology for Biochemists**. *Lectures*. Course no. 23700. Fall. 2017.

[7] **Botany & Plant Physiology for Veterinary Sciences**. *Lectures*. Course no. 23760b-c. Fall. 2017.

[6] **Introduction to Botany & Biodiversity**. *Lectures*. Course no. 23106. Fall. 2017.

[5] **Introduction to Botany & Biodiversity**. *Laboratories*. Course no. 23108a-e. Fall. 2017.

[4] **Topics in Biology: Evolution**. *Graduate seminar*. Course no. 23653. Spring. 2017.

[3] **Topics in Biology: Evolution**. *Graduate laboratories*. Course no. 23654a-b. Spring. 2017.

[2] **Introduction to Botany & Biodiversity**. *Laboratories*. Course no. 23108a-e. Fall. 2016.

[1] **Evolution & Biodiversity–Botany**. *Graduate lectures*. Course no. 23303a-e. Spring. 2015.

Student Supervision

Graduate Students

[17] STUDENT: Louisa Acquah. ROLE: Primary thesis advisor. Fort Hays State University, 2023–2026.

[16] STUDENT: David Esteban Bohorquez. ROLE: Primary thesis advisor. Fort Hays State University, 2025–ongoing.

[15] STUDENT: Eka Giorgashvili. ROLE: Primary thesis advisor. Freie Universität Berlin, 2019–2020.

[14] STUDENT: Thanina Hamitouche. ROLE: Primary thesis advisor. Fort Hays State University, 2025–ongoing.

[13] STUDENT: Yannick Hartmaring. ROLE: Primary thesis advisor. Freie Universität Berlin, 2020–2021.

[12] STUDENT: Nils Jenke. ROLE: Primary thesis advisor. Freie Universität Berlin, 2020–2021.

[11] STUDENT: Buddha Thapa Magar. ROLE: Primary thesis advisor. Fort Hays State University, 2024–ongoing.

[10] STUDENT: Jessica Röstel. ROLE: Primary thesis advisor. Freie Universität Berlin, 2019–2020.

Undergraduate Students

[9] STUDENT: Elijah McCullough. ROLE: Capstone project advisor. Fort Hays State University, 2025–2026.

[8] STUDENT: Lauren Perez. ROLE: Capstone project advisor. Fort Hays State University, 2026–ongoing.

[7] STUDENT: Bryce Steffan. ROLE: Capstone project advisor. Fort Hays State University, 2025–2026.

[6] STUDENT: Brandon Williams. ROLE: Capstone project advisor. Fort Hays State University, 2025.

[5] STUDENT: Gregory M. Smith. ROLE: Capstone project advisor. Fort Hays State University, 2024.

[4] STUDENT: Phongsavanh Mongkhonvilay. ROLE: Capstone project advisor. Fort Hays State University, 2023.

[3] STUDENT: Tilman Mehl. ROLE: Thesis advisor. Freie Universität Berlin, 2020.

[2] STUDENT: Yannick Hartmaring. ROLE: Thesis advisor. Freie Universität Berlin, 2019.

[1] STUDENT: Nils Jenke. ROLE: Thesis advisor. Freie Universität Berlin, 2019.

Honors, Awards & Certificates for Teaching

Certificate in Effective Teaching Practices — Assoc. of College and Univ. Educators — 2025

9-month (25-module) course in effective teaching practices on implementation of evidence-based instructional approaches

Teaching grant–Experiential learning innovation — Fort Hays State University — 2023

Teaching grant–Undergraduate research experience — Fort Hays State University — 2023

Teaching grant–Industry 4.0 — Freie Universität Berlin — 2018

Teaching award — Freie Universität Berlin — 2017

Teaching assistant award — University of Texas at Austin — 2007

Service

Leadership Training

LHH Executive Program for Leaders & Managers — LHH | OTM Career Development — 2022

4-month training in communication, transition management, interviews, hearings, and assessments

Committee work

University committees

Strategic Planning & Improv. Committee, Chair	Fort Hays State University	2025–ongoing
Faculty Senate, Full Member	Fort Hays State University	2025–ongoing
Departm. Hiring Committee, Chair	Fort Hays State University	2025
Faculty Senate, Alternate Member	Fort Hays State University	2024–2025
Departm. Graduate Education Committee, Member	Fort Hays State University	2024–ongoing
Departm. Hiring Committee, Member	Fort Hays State University	2024
Departm. Scholarship Committee, Member	Fort Hays State University	2024
Departm. Infrastructure Committee, Member	Freie Universität Berlin	2017–2018

Peer-review

Funding Agencies

- Deutsche Forschungsgemeinschaft (DFG)

Scientific Journals

- Annals of Botany
- BMC Plant Biology
- Botanical Journal of the Linnean Society
- Frontiers in Plant Science
- GigaScience
- Mathematical Biosciences
- Mitochondrial DNA Part B
- Molecular Ecology
- Molecular Ecology Resources
- Nordic Journal of Botany
- Plant Systematics and Evolution
- PLOS One
- Systematic Botany
- Taxon
- Willdenowia

Scientific Memberships

- International Society for Computational Biology (ISCB)
- International Association for Plant Taxonomy (IAPT)
- Austrian Scientists & Scholars in North America (ASCINA)
- German Association of University Professors and Lecturers (DHV)
- German Botanical Society (DBG)
- Council on Undergraduate Research (CUR)

Last updated: 2026-06-20